Topic: The heat equation

Question: Find a solution to the partial differential equation.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \text{ for } 0 \leq x \leq 10$$

$$u(x,0) = 2\sin(\pi x) + 4\sin(2\pi x)$$

$$u(0,t) = u(10,t) = 0$$

Answer choices:

A $u(x,t) = 2\sin(\pi x)e^{4\pi^2 t} + 4\sin(2\pi x)e^{-16\pi^2 t}$

B $u(x,t) = 2\cos(\pi x)e^{4\pi^2 t} + 4\cos(2\pi x)e^{-16\pi^2 t}$

C $u(x,t) = 2\sin(\pi x)e^{4\pi^2 t} + 4\sin(2\pi x)e^{-8\pi^2 t}$

D $u(x,t) = 2\cos(\pi x)e^{4\pi^2 t} + 4\cos(2\pi x)e^{-8\pi^2 t}$

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Solution: A

If we start with the product solution $u(x,t) = v(x)w(t)$, then we know

$$v(x)w'(t) = 4v''(x)w(t)$$

$$\frac{w'}{4w}(t) = \frac{v''}{v}(x) = -\lambda$$

Break this equation into two ordinary differential equations.

$$w' = -4\lambda w \qquad\qquad v'' = -\lambda v$$

The solution to the first equation, which is a linear differential equation, is $w(t) = Ce^{-4\lambda t}$. To solve the second equation, we find the associated characteristic equation.

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$ and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v(0) = c_1 + c_2 = 0$$

$$v(10) = c_1 e^{-10\sqrt{-\lambda}} + c_2 e^{10\sqrt{-\lambda}} = 0$$

which allows us to find $c_1 = c_2 = 0$, from which we get only the trivial solution. If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$ and the solution is

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$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v(10) = c_1 + 10c_2 = 0$$

which allows us to find $c_1 = c_2 = 0$, from which we again get only the trivial solution. If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$ and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v(10) = c_1 \cos(10\sqrt{\lambda}) + c_2 \sin(10\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution when λ satisfies

$$\sin(10\sqrt{\lambda}) = 0$$

$$10\sqrt{\lambda} = \pi n$$

$$\lambda = \left(\frac{\pi n}{10}\right)^2$$

So the general solution equation for the $\lambda > 0$ case becomes

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$$v(x) = (0)\cos\left(\sqrt{\left(\frac{n\pi}{10}\right)^2}\,x\right) + c_2 \sin\left(\sqrt{\left(\frac{n\pi}{10}\right)^2}\,x\right)$$

$$v(x) = c_2 \sin\left(\frac{n\pi x}{10}\right) \qquad \text{with } n = 1, 2, 3, \ldots$$

The first order ordinary differential equation is a linear equation in standard form,

$$w' + k\lambda w = 0$$

so we know the solution is

$$w = Ce^{-4\lambda t}$$

$$w = Ce^{-4\left(\frac{n\pi}{10}\right)^2 t}$$

Putting our results together from the first and second order equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u_n(x,t) = c_2 \sin\left(\frac{n\pi x}{10}\right) Ce^{-4\left(\frac{n\pi}{10}\right)^2 t} \qquad \text{with } n = 1, 2, 3, \ldots$$

While c_2 will be different for each n, the constant $C \times c_2$ will also depend on n, so let's rename it as B_n.

$$u_n(x,t) = \sum_{n=0}^{\infty} B_n \sin\left(\frac{n\pi x}{10}\right) e^{-4\left(\frac{n\pi}{10}\right)^2 t} \qquad \text{with } n = 1, 2, 3, \ldots$$

Matching up the boundary condition

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$$u(x,0) = 2\sin(\pi x) + 4\sin(2\pi x)$$

to the general solution of the heat equation, we can identify $n = 10$ and therefore $B_n = B_{10} = 2$ for the first term, and $n = 20$ and therefore $B_n = B_{20} = 4$ for the second term. Which means that, given this initial condition, the solution to the heat equation would be

$$u(x,t) = u_{10}(x,t) + u_{20}(x,t)$$

$$u(x,t) = 2\sin(\pi x)e^{-4\pi^2 t} + 4\sin(2\pi x)e^{-16\pi^2 t}$$

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Topic: The heat equation

Question: Find a solution to the partial differential equation.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \text{ for } 0 \leq x \leq 5$$

$$u(x,0) = 10 + \cos(\pi x)$$

$$u_x(0,t) = u_x(5,t) = 0$$

Answer choices:

A $u(x,t) = 10 + \sin(\pi x)e^{-\pi^2 t}$

B $u(x,t) = 10 + \cos(\pi x)e^{-\pi^2 t}$

C $u(x,t) = 10e^{-\pi^2 t} + \sin(\pi x)e^{-\pi^2 t}$

D $u(x,t) = 10e^{-\pi^2 t} + \cos(\pi x)e^{-\pi^2 t}$

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Solution: B

If we start with the product solution $u(x,t) = v(x)w(t)$, then we know

$$v(x)w'(t) = v''(x)w(t)$$

$$\frac{w'}{w}(t) = \frac{v''}{v}(x) = -\lambda$$

Break this equation into two ordinary differential equations.

$$w' = -\lambda w \qquad\qquad v'' = -\lambda v$$

The solution to the first equation, which is a linear differential equation, is $w(t) = Ce^{-\lambda t}$. To solve the second equation, we find the associated characteristic equation.

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$ and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v'(0) = -c_1 + c_2 = 0$$

$$v'(5) = -c_1\sqrt{-\lambda}e^{-5\sqrt{-\lambda}} + c_2\sqrt{-\lambda}e^{5\sqrt{-\lambda}} = 0$$

which allows us to find $c_1 = c_2 = 0$, from which we get only the trivial solution. If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$ and the solution is

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$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

$$v'(0) = c_2 = 0$$

$$v'(5) = c_2 = 0$$

which allows us to find $v(x) = c_1$, a non-trivial solution. If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$ and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives

$$v'(0) = c_2 = 0$$

$$v'(5) = -c_1\sqrt{\lambda}\sin(5\sqrt{\lambda}) + c_2\sqrt{\lambda}\cos(5\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution when λ satisfies

$$\sin(5\sqrt{\lambda}) = 0$$

$$5\sqrt{\lambda} = \pi n$$

$$\lambda = \left(\frac{\pi n}{5}\right)^2$$

So the general solution equation for the $\lambda > 0$ case becomes

$$v(x) = c_1 \cos\left(\sqrt{\left(\frac{n\pi}{5}\right)^2}x\right) + (0)\sin\left(\sqrt{\left(\frac{n\pi}{5}\right)^2}x\right)$$

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$$v(x) = c_1 \cos\left(\frac{n\pi x}{5}\right) \qquad \text{with } n = 1, 2, 3, \ldots$$

The first order ordinary differential equation is a linear equation in standard form,

$$w' + k\lambda w = 0$$

so we know the solution is

$$w = Ce^{-\lambda t}$$

$$w = Ce^{-\left(\frac{n\pi}{5}\right)^2 t}$$

Putting our results together from the first and second order equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u_n(x,t) = \sum_{n=0}^{\infty} B_n \cos\left(\frac{n\pi x}{5}\right) e^{-\left(\frac{n\pi}{5}\right)^2 t} \qquad \text{with } n = 1, 2, 3, \ldots$$

We can rewrite

$$u_n(x,t) = B_0 \cos 0 e^{-(0)t} + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi x}{5}\right) e^{-\left(\frac{n\pi}{5}\right)^2 t}$$

$$u_n(x,t) = B_0 + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi x}{5}\right) e^{-\left(\frac{n\pi}{5}\right)^2 t}$$

Matching this up to the boundary condition

$$u(x,0) = 10 + \cos(\pi x)$$

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gives

$$10 + \cos(\pi x) = B_0 + \sum_{n=1}^{\infty} B_n \cos\left(\frac{n\pi x}{5}\right) e^{-\left(\frac{n\pi}{5}\right)^2 t}$$

So we have $B_0 = 10$ and $B_5 = 1$.

$$u(x,t) = u_0(x,t) + u_5(x,t)$$

$$u(x,t) = 10 + \cos(\pi x) e^{-\pi^2 t}$$

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Topic: The heat equation

Question: Find a solution to the partial differential equation.

$$\frac{\partial u}{\partial t} = 9\frac{\partial^2 u}{\partial x^2} \text{ for } 0 \leq x \leq \pi$$

$$u(x,0) = 3\sin\left(\frac{x}{2}\right)$$

$$u(0,t) = u_x(\pi, t) = 0$$

Answer choices:

A $u(x,t) = 3\sin\left(\frac{1}{2}x\right)e^{-\frac{9}{2}t}$

B $u(x,t) = 3\cos\left(\frac{1}{2}x\right)e^{-\frac{9}{2}t}$

C $u(x,t) = 3\sin\left(\frac{1}{2}x\right)e^{-\frac{9}{4}t}$

D $u(x,t) = 3\cos\left(\frac{1}{2}x\right)e^{-\frac{9}{4}t}$

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Solution: C

If we start with the product solution $u(x,t) = v(x)w(t)$, then we know

$$v(x)w'(t) = 9v''(x)w(t)$$

$$\frac{w'}{9w}(t) = \frac{v''}{v}(x) = -\lambda$$

Break this equation into two ordinary differential equations.

$$w' = -9\lambda w \qquad\qquad v'' = -\lambda v$$

The solution to the first equation, which is a linear differential equation, is $w(t) = Ce^{-9\lambda t}$. To solve the second equation, we find the associated characteristic equation.

$$r^2 + \lambda = 0$$

From the characteristic equation, if $\lambda < 0$, the equation has distinct real roots $r = \pm\sqrt{-\lambda}$ and the solution is

$$v(x) = c_1 e^{-\sqrt{-\lambda}x} + c_2 e^{\sqrt{-\lambda}x}$$

Applying the boundary conditions gives

$$v(0) = c_1 + c_2 = 0$$

$$v'(\pi) = -c_1\sqrt{-\lambda}e^{-\pi\sqrt{-\lambda}} + c_2\sqrt{-\lambda}e^{\pi\sqrt{-\lambda}} = 0$$

which allows us to find $c_1 = c_2 = 0$, from which we get only the trivial solution. If $\lambda = 0$, the characteristic equation has equal real roots $r = 0$ and the solution is

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$$v(x) = c_1 + c_2 x$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v'(\pi) = c_2 = 0$$

which allows us to find $c_1 = c_2 = 0$, from which we again get only the trivial solution. If $\lambda > 0$, the characteristic equation has complex roots $r = \pm\sqrt{\lambda}i$ and the solution is

$$v(x) = c_1 \cos(\sqrt{\lambda}x) + c_2 \sin(\sqrt{\lambda}x)$$

Applying the boundary conditions gives

$$v(0) = c_1 = 0$$

$$v'(\pi) = -c_1\sqrt{\lambda}\sin(\pi\sqrt{\lambda}) + c_2\sqrt{\lambda}\cos(\pi\sqrt{\lambda}) = 0$$

We'll find a non-trivial solution when λ satisfies

$$\cos(\pi\sqrt{\lambda}) = 0$$

$$\pi\sqrt{\lambda} = \frac{\pi}{2} + \pi n$$

$$\lambda = \left(\frac{1}{2} + n\right)^2$$

So the general solution equation for the $\lambda > 0$ case becomes

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$$v(x) = (0)\cos\left(\sqrt{\left(\frac{n\pi}{5}\right)^2}\,x\right) + c_2 \sin\left(\sqrt{\left(\frac{n\pi}{5}\right)^2}\,x\right)$$

$$v(x) = c_2 \sin\left(\left(\frac{1}{2} + n\right)x\right) \qquad \text{with } n = 1, 2, 3, \ldots$$

Plugging the value of λ we found into the solution equation for w, we get

$$w(t) = Ce^{-9\lambda t}$$

$$w = Ce^{-9\left(\frac{1}{2}+n\right)^2 t}$$

Putting our results together from the first and second order equations, we get the product solution to the heat equation.

$$u(x,t) = v(x)w(t)$$

$$u_n(x,t) = \sum_{n=0}^{\infty} B_n \sin\left(\left(\frac{1}{2} + n\right)x\right)e^{-9\left(\frac{1}{2}+n\right)^2 t} \qquad \text{with } n = 1, 2, 3, \ldots$$

Matching this up to the boundary condition

$$u(x,0) = 3\sin\left(\frac{x}{2}\right)$$

gives

$$\sum_{n=0}^{\infty} B_n \sin\left(\left(\frac{1}{2} + n\right)x\right)e^{-9\left(\frac{1}{2}+n\right)^2 t} = 3\sin\left(\frac{x}{2}\right)$$

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So we have $B_0 = 3$ and $B_1 = B_2 = \ldots = 0$.

$$u(x,t) = 3\sin\left(\frac{1}{2}x\right)e^{-\frac{9}{4}t}$$

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